

Homework 9 Key

November 2025

Problem 1

There can be four different scenarios that a committee can contain at least 3 women.

1. **3 women and 3 men:**

$$\binom{8}{3} \times \binom{10}{3} = \frac{8 \times 7 \times 6}{3 \times 2 \times 1} \times \frac{10 \times 9 \times 8}{3 \times 2 \times 1} = 6720$$

2. **4 women and 2 men:**

$$\binom{8}{4} \times \binom{10}{2} = \frac{8 \times 7 \times 6 \times 5}{4 \times 3 \times 2 \times 1} \times \frac{10 \times 9}{2 \times 1} = 3150$$

3. **5 women and 1 men:**

$$\binom{8}{5} \times \binom{10}{1} = \binom{8}{3} \times 10 = \frac{8 \times 7 \times 6}{3 \times 2 \times 1} \times 10 = 560$$

4. **6 women:**

$$\binom{8}{6} \times \binom{10}{0} = \binom{8}{2} \times 1 = \frac{8 \times 7}{2 \times 1} \times 1 = 28$$

Therefore, there can be $6720 + 3150 + 560 + 28 = 10458$ different committees.

Problem 2

To include a specific book, B , in an ordered list, first choose 3 books from 11 distinct books other than the book B . Then, make an ordered reading list.

1. **Pick 3 books from 11 distinct books:**

$$\binom{11}{3} = \frac{11 \times 10 \times 9}{3 \times 2 \times 1} = 165$$

2. Feasible ordered lists given a combination from the previous step:

Make ordered lists of four books (three from the previous step and the book B).

$$4! = 24$$

3. All possible ordered lists:

$$165 \times 24 = 3960$$

Problem 3

For each digit, we can use 1, 2, 3, 4, 5, 6 and repetition is allowed. Hence, there can be $6^3 = 216$ different numbers for the first three digits. Since each number must be even, we can use 2, 4, 6 for the last digit. Therefore, there can be $216 \times 3 = 648$ different numbers.

Problem 4

We have 9 distinct letters: P, R, O, B, A, I, L, T, Y. Hence, the number of possible arrangements is:

$$\frac{11!}{2!2!}$$

Since we must not double-count the permutation of B and I, we divide by $2!$ for each letter. If we consider cases where O, A, I, I are all together, the possible cases of this block is $4!/2! = 12$ since we must not double-count the permutation of I. Hence, the number of arrangements is $8!/2! \times 12$ when we consider the block of O, A, I, I a letter. Therefore, the probability is:

$$\frac{\frac{8!}{2!} \times 12}{\frac{11!}{2!2!}} = \frac{8! \times 12}{\frac{11!}{2!}} = \frac{8! \times 24}{11!} = \frac{24}{11 \times 10 \times 9} = \frac{4}{11 \times 5 \times 3} = \frac{4}{165}$$

Problem 5

For a number to be a 3-digit number divisible by 5 without repetition, the last digit must be 5. Hence, we can pick two numbers from $1, \dots, 4, 6, \dots, 9$ for the first two digits:

$${}^8P_2 = 8 \times 7$$

- **Without repetition:** Total number of possible 3-digit numbers is

$${}^9P_3 = 9 \times 8 \times 7$$

Therefore, the probability is

$$\frac{8 \times 7}{9 \times 8 \times 7} = \frac{1}{9}$$

- **With repetition:** Total number of possible 3-digit numbers is

$$9^3 = 729$$

Therefore, the probability is

$$\frac{8 \times 7}{9^3} = \frac{56}{729}$$

Problem 6

To win 5 times, the player must play at least 5 matches.

1. Probability of finishing on the 5th match:

The probability of winning the first 5 matches in a row is

$$P(X = 4 \cap n = 4) \times p = \underbrace{\binom{4}{4} \cdot p^4 \cdot (1-p)^{4-4}}_{\text{probability of } 4-0} \cdot p = p^5$$

2. Probability of finishing on the 6th match:

$$P(X = 4 \cap n = 5) \times p = \underbrace{\binom{5}{4} \cdot p^4 \cdot (1-p)^{5-4}}_{\text{probability of } 4-1} \cdot p$$

,
where X is the number of wins and n is the number of matches before the last match.

3. Probability of finishing on the 7th match:

$$P(X = 4 \cap n = 6) \times p = \underbrace{\binom{6}{4} \cdot p^4 \cdot (1-p)^{6-4}}_{\text{probability of } 4-2} \cdot p$$

4. Probability of finishing on the 8th match:

$$P(X = 4 \cap n = 7) \times p = \underbrace{\binom{7}{4} \cdot p^4 \cdot (1-p)^{7-4}}_{\text{probability of } 4-3} \cdot p$$

5. Probability of finishing on the 9th match:

$$P(X = 4 \cap n = 8) \times p = \underbrace{\binom{8}{4} \cdot p^4 \cdot (1-p)^{8-4}}_{\text{probability of } 4-4} \cdot p$$

Therefore the probability is:

$$\begin{aligned} & P(X = 4 \cap (4 \leq n \leq 8)) \\ &= p^5 + \frac{5!}{4!1!} p^4 \cdot (1-p)^{5-4} \cdot p + \frac{6!}{4!2!} p^4 \cdot (1-p)^{6-4} \cdot p + \frac{7!}{4!3!} p^4 \cdot (1-p)^{7-4} \cdot p + \frac{8!}{4!4!} p^4 \cdot (1-p)^{8-4} \cdot p \\ &= p^5 (1 + 5(1-p) + 15(1-p)^2 + 35(1-p)^3 + 70(1-p)^4) \\ &= p^5 (1 + 5(1-p)(1 + 3(1-p)) + 35(1-p)^3(1 + 2(1-p))) \\ &= p^5 (1 + 5(1-p)(4-3p) + 35(1-p)^3(3-2p)) \\ &= p^5 (1 + 5(1-p)((4-3p) + 7(1-p)^2(3-2p))) \end{aligned}$$

For $p = 0.6$,

$$\begin{aligned} &= 0.6^5 (1 + 5 \times 0.4 \times (2.2 + 7 \times 0.4^2 \times 1.8)) \\ &= 0.6^5 (1 + 2 \times (2.2 + 2.016)) \\ &= 0.6^5 (1 + 8.432) \\ &= 0.6^5 \times 9.432 \\ &= 0.73343232 \end{aligned}$$

Problem 7

$$\begin{aligned} P(X \geq 4) &= \underbrace{\left[\binom{5}{4} \times 0.7^4 \times 0.3^1 \right]}_{P(X=4)} + \underbrace{\left[\binom{5}{5} \times 0.7^5 \times 0.3^0 \right]}_{P(X=5)} \\ &= 5 \times 0.7^4 \times 0.3 + 0.7^5 \\ &= 0.7^4 (1.5 + 0.7) \\ &= 0.7^4 \times 2.2 \\ &= 0.52822 \end{aligned}$$